

DENTFLUENCE V1 — FINAL IMPLEMENTATION READINESS REPORT

Tulip Dental Production Deployment Assessment · Third Independent Audit

Date: 7 August 2026 · Scope: **Can Tulip Dental switch off every old system tomorrow at 8:00 AM and run only on Dentfluence?** · Method: full workflow trace through blade views, controllers, services, observers, scheduled commands, routes and API — two independent code-tracing passes on top of the two prior audits (architecture 08-03, production readiness 08-07 AM). Every claim below is file-evidenced; anything unverifiable from the repository is marked as such.



SECTION 1 — CAN TULIP DENTAL GO LIVE?

⚖️ YES WITH CONDITIONS

Yes, because the clinical-financial spine holds end to end. This audit walked the actual chain in code: registration (6 required fields, 30-second modal with duplicate-guard), booking (global modal with quick-register and walk-in tabs), consultation → one-click treatment plan (`treatment-plans.from-consultation`), a properly modelled present/accept/defer/reject/revert decision flow, a visit form that auto-loads plan items into a worksheet, a billing prompt that pre-fills the invoice from the visit, a payment modal covering all 9 Indian payment modes with wallet allocation, receipt generation, a full Rx pad with print/PDF/WhatsApp routes, a real cashbook for the evening cash tally, and a purpose-built **competitor migration importer with Clinicia and Bestosys column profiles** for day-1 patient cutover. No step in the money or clinical path is missing.

Conditions, because two classes of issues remain. First, the 4 P0 + 7 P1 items from the morning production audit stand (no backups, AI RBAC bypass, public consultation photos, zero error visibility, uncommitted sprint, env hardening — all closable in ~7 days). Second, this audit found the **go-live configuration is currently a no-op**: `.env` has WhatsApp in dry-run with blank credentials and `MAIL_MAILER=log`, which silently disables every reminder, review request, morning briefing and evening summary. The clinic can operate manually on day 1 — but the automation layer the staff will be told exists **does not fire until three env values change**.

What it is not: it is not blocked by missing features. Every friction point found (Section 7) has a manual workaround a trained receptionist can absorb.

SECTION 2 — DAY IN THE LIFE SIMULATION (traced in code, not assumed)

08:00 OWNER opens Dashboard → **WORKS**: 4 KPI cards (waiting/in-chair/done, collected today, outstanding, pending lab), alert strip, Today's Schedule with per-row patient links, link to /huddle web board. **FRICTION**: the CLI huddle (tulip:huddle 08:00) writes to storage/logs/daily-huddle.log – a briefing nobody can read; the web /huddle board is a separate implementation. Morning-briefing EMAIL goes to log (MAIL_MAILER=log).

08:15 PATIENT CALLS → RECEPTION books → **WORKS via global modal**: typeahead, quick-register inline, duplicate-phone warning, walk-in tab. **TRAP**: the standalone Book Appointment PAGE cannot take a new patient – "Create new patient" navigates away and discards the half-typed booking. Train staff on the modal only.

08:30 NEW PATIENT arrives → REGISTRATION → **WORKS**: 5-tab modal, 6 required fields, "Save & skip remaining", paper-form photo scan prefill, family-link duplicate flow with 3 exits.

09:00 CHECK-IN → **PARTIAL**: a live queue page (/appointments/today: 30-s auto-refresh, chair utilization, Check In / Start / Done / No-Show) EXISTS but is **orphaned – linked from no menu**. Reception must know the URL.

09:10 DOCTOR starts consultation → **GAP**: no "Start Consultation" on any appointment. Path is appointment → patient profile → Consultation tab → New Consultation (3 screens, chief complaint re-typed). The appointment_id link is an optional dropdown buried in the backdating strip – so the appointment→consultation wiring observer **almost never fires**, the appointment stays "in chair", and someone must press Done manually.

09:40 CONSULTATION done → TREATMENT PLAN → **WORKS – best handoff in the app**: one violet button, prefilled from the consultation.

10:00 PLAN presented chairside → **WORKS**: Present/Estimate → Accept (gated on presented) / Defer / Reject / Revert-with-reason, all inline AJAX; print with patient QR; consent form picker.

10:20 TREATMENT VISIT → **PARTIAL**: accepted plan card has "Bill from Plan" but **no "Start Visit" button** – switch to Visits tab, Add Visit, re-select plan from dropdown; then the worksheet auto-loads plan items (good). The post-consultation visit gate exists but only fires for appointment type = 'treatment'.

11:00 VISIT saved → BILLING → **PARTIAL**: save is silent JSON; a BillingPrompt is queued and surfaces as an amber banner on the patient's Billing tab + a sidebar badge (30-s poll). "Build Invoice" pre-fills items from the visit – **no re-picking** – but nothing actively tells the desk the patient is walking over with money owed.

11:10 PAYMENT → **WORKS**: in-page modal, 9 modes with mode-conditional fields, wallet/advance allocation, "₹X recorded. Receipt N generated." **FRICTION**: receipt print is not one click – back on the invoice, find the payment row, click the receipt number, new tab. No WhatsApp button

anywhere in billing views.

11:30 PRESCRIPTION → **PARTIAL**: full Rx pad with print?auto=1 and PDF – but the consultation's "Rx" button routes to a profile TAB, not the pad: 3 screens from the chair.

12:00 LAB work needed → **WORKS – automatic**: the visit's lab section spawns a draft LabCase inheriting teeth; 9-status lifecycle via one transition service (web + API); overdue cases become urgent tasks daily at 09:00; on final_received a "call patient in" task fires. Monthly vendor reconciliation screen exists (draft→review→approved/disputed).

13:00 CHECKOUT → **GAP**: nothing prompts "book next appointment / set recall". The visit form's next_visit_date/type quick-chips are collected and **read by nothing on the server** – display-only theatre. Recall is entirely the background engine keyed on last_visit_date.

16:00 WHATSAPP → **PARTIAL**: desk sends are wa.me click-to-chat links (patient header, calendar queue) – staff's own phone, interim by design; Rx has a real API send route. Automated reminders (10:00) and review requests (11:00) are coded and consent-gated but currently **dry-run no-ops**.

18:00 END OF DAY → **GAP**: no day-close screen. Pieces exist – Finance dashboard cash-in-hand, Cashbook (day-wise in/out/net/running balance), payment-mode breakdown ("real data, not a mock split"), Income tabs – but the receptionist tours 3-4 screens to close the till. Evening summary is an email (currently to log).

Bottom line of the simulation: every workflow completes; six of them complete only because a human remembers the next step. That is operable — clinics ran on paper — but Week-1 training must cover exactly these six hand-offs (Section 7 list).

SECTION 3 — ROLE READINESS

ROLE	ASSESSMENT (EVIDENCE-BASED)	SCORE
Owner	Can run the clinic. 6-tab Reports (appointments, revenue, patients, treatment, lab, inventory) with day/mode granularity, KPI dashboard, huddle report, cashbook, GST + CA Export for the accountant. Gaps: no revenue-per-doctor anywhere in ReportsController (appointment counts per doctor only) — the first report an owner asks for; analytics snapshots job is shadow-only.	85
Manager	Can monitor: web huddle board (schedule, safety flags, money to collect, opportunities, no-shows, tasks, lab, inventory alerts, new patients), accountability view, SLA alert email at 14:00, task engine with shift reminders every 5 min. Gaps: briefings are email-only (currently log-only); no single end-of-day console.	82
Front Desk	Can work a full day: booking modal with quick-register + walk-in, 30-second registration, check-in queue (once linked in nav), billing prompt → prefilled invoice, 9-mode payment modal, wallet handling. Carries the burden of the six manual hand-offs: pressing Done on appointments, watching the billing badge, prompting rebooking at checkout, sending wa.me messages from a phone.	80
Assistant / Associate Doctor	Can work independently: consultation (5 types), one-click plan creation, present/accept chairside, visit worksheet from plan items, Rx pad with drug master + pediatric ml dosing + consent gate, lab case auto-spawned from visit. Friction: 3 screens to reach the consultation form and 3 to the Rx pad; no chairside "start visit" from the plan.	84
Accounts	Strongest role. Transactional locked billing, cashbook, real payment-mode split, expense entry with OCR bill scan, vendors, banking, vouchers, GST view, CA export, wallet register, monthly lab reconciliation workflow. Payroll is a register (list/create/delete) — not a payroll engine; fine for one clinic.	92
Lab coordinator	Operable: 3 creation surfaces, 9-status flow through one transition service, auto-tasks on overdue, auto comm-close on terminal states, reconciliation. Attachment privacy is a P1 (public disk).	85
Marketing	Blog Hub is live and complete (CRUD, versioning, publish). Reviews module coded, gated on live WhatsApp. GBP/Meta integrations parked by decision. Adequate for V1's single clinic; not a growth engine yet.	70

SECTION 4 — ANDROID READINESS

Declaration: No Flutter/Dart source exists in this repository (verified: zero `pubspec.yaml`, zero `.dart` files; only a roadmap document). The Flutter app itself is **not verifiable from the available code**. This section therefore audits the API contract the app consumes (fully verifiable) plus the governance record of app testing.

The API is not the bottleneck. 28 Api/V1 controllers, ~125 write routes, Sanctum auth, per-route

`api.role:module:x,action` RBAC (stricter than web), and full write parity on the five critical flows: create appointment (incl. walk-in, block-slot, reschedule, status), create consultation (all 4 typed variants + COHA), record payment (incl. EMI, wallet, discounts, void), create/edit visit, prescriptions (incl. finalize/repeat/alerts). Even treatment-plan present/accept/defer/partial-accept and invoice creation are present. API gaps: finance module entirely (expenses/payroll/cashbook/GST), patient import, lab reconciliation, HR, settings.

USER ON ANDROID ONLY	CLASSIFICATION	WHY
Doctor (read + light write)	Needs Small Work	API supports the full clinical day; the app's write paths are code-complete but untested per governance record (only Membership confirmed tested). One smoke pass per flow away from "Ready".
Assistant	Needs Small Work	Same basis — visit/consultation writes exist server-side, unvalidated client-side.
Owner	Needs Small Work	Dashboard/Report API controllers exist (read-only by design). Usable for monitoring once app read paths are smoke-tested.
Manager	Needs Small Work	Huddle API is rich (yesterday-flow, comms-push, task assign). Untested client.
Receptionist	Not Ready	Desk work is dense multi-window work (calendar + patient + billing simultaneously); no finance API; the desk stays on desktop for V1 by design.

V1 rule (restated from audit #2): Android ships as a read-first companion; every mobile write path earns production status only after one supervised smoke test. All polish → V1.1. **Android readiness: 58/100** (API ~85, app unverifiable/untested ~45, weighted).

SECTION 5 — AUTOMATION READINESS

22 scheduled jobs verified in `routes/console.php`, all `withoutOverlapping()`. Design philosophy is consistent and deliberate: **automation creates tasks for humans; only consented, transactional messages auto-send**. That is the right call for an Indian clinic — but today the auto-send half is disabled by config.

AUTOMATION	CLASS	EVIDENCE / CAVEAT
Appointment reminders (calls)	Semi-automatic	08:00 job creates "Reminder call" tasks for reception — no message sent, by design.
Appointment reminders (WhatsApp)	Fully automatic*	10:00 job sends Meta template with Confirm/Reschedule/Cancel buttons, dedup-keyed, consent-gated. *Dry-run no-op today ; blocked sends are counted only in a logfile — no staff-facing "blocked" report.
Recall	Semi-automatic	07:00 engine, 6 triggers → communication_queue for staff action; service docblock: "no automatic send pipeline". Correct for V1; the queue screen needs pagination (P1) and the 1,810 backlog needs a cap.
Payment overdue	Semi-automatic	07:15 scan → payment.overdue event → follow-up task (≥₹500, day-3).
Review requests	Fully automatic*	11:00, one-per-appointment-ever, targets yesterday's done appointments. *Gated on live WhatsApp.
Birthdays	Semi-automatic	Explicit design: emits event → staff task; "a human decides whether to send". 360-day cooldown.
Lab overdue → tasks	Fully automatic	09:00; urgent task to manager/admin; trial-untouched-2-days → doctor review task; dedup via active_task_id.
Daily huddle	Automatic but undelivered	Generation automatic; scheduled output goes to a logfile. The web /huddle board is the real delivery — link it and delete the log ritual.
Briefings / SLA / evening summary	Fully automatic*	07:05 / 14:00 / 18:00 — all EMAIL (Mail::to), no in-app fallback. *Dead while MAIL_MAILER=log.
Membership expiry	Semi-automatic	30-days-out scan → renewal call task, 25-day cooldown.
Notifications (in-app)	Automatic	Producers: lab events (richest), task assignment, shift reminders (5-min), periodic task nudges (2-h), audit-verify failure → all admins. 30-s sidebar polling.
Inventory low-stock	Manual	The one genuinely missing automation. No scheduled job, no observer, no notification — status computed on-read; surfaces only in huddle + reports. A stockout will happen unless someone looks.

Journey engine	Semi-automatic	ActivityEngine events → RulesEngine → TaskEngine chain is live (overdue, birthday, lab-ready, membership, escalation ₹30k/2h). Fires tasks, not messages.
Reports	Automatic	Projections rebuilt every 15 min (today + analytics; analytics snapshot is shadow-only for now).

Automation readiness 78/100: the engine is real and thoughtfully gated, minus points because its two delivery channels (WhatsApp, mail) are dead in current config, the huddle ships to a logfile, and inventory has nothing.

SECTION 6 — AI READINESS

CAPABILITY	RATING	ASSESSMENT
Assistant "Tulip" (current)	Disable for V1	Real tool-calling architecture (11 tools, confirm-gated writes, action log) — but <code>runTool()</code> has zero permission checks (re-verified twice). 25/100 as shipped. <code>ASSISTANT_ENABLED=false</code> is a go-live condition.
Vision form-scan (current)	V1 — keep	Registration "scan paper form" prefill and expense bill OCR (ReceiptScanService) are shipped, scoped, low-risk. The quiet AI wins.
Voice notes / local AI (ollama, whisper)	V1.1	Config present (private-disk PHI note is correct); depends on local models; not part of daily core.
Clinical AI / decision support	V2	Rx alert-check endpoints exist as rules, not AI. Anything generative-clinical needs the authorization layer plus a validation protocol first.
Reception AI (auto-reply, triage)	V2	Chatbot webhook controller exists unrouted. Build only after WhatsApp live traffic exists to learn from.

Sequencing recommendation: V1 = vision prefill only. V1.1 = re-enable Tulip behind per-tool `canAccess()` + read-only. V2 = write tools, voice, reception AI. This matches CEO Directive #004 exactly.

SECTION 7 — WORKFLOW GAPS ("What should I do next?" points)

GAP (ALL CODE-VERIFIED)	SEVERITY	DISPOSITION
1. <code>/appointments/today</code> live queue orphaned — linked from no menu	High	☐ FIX BEFORE SHIP — one sidebar/dashboard link. Minutes of work, unlocks the best screen for the desk.
2. No "Start Consultation" from an appointment; <code>appointment_id</code> never set → appointments never auto-close	High	☐ FIX BEFORE SHIP (minimal form) — add one "Start Consultation" link on the today-queue card passing <code>appointment_id</code> ; the observer already does the rest. Not a redesign — one link + one prefill. Full flow polish → V1.1.
3. Standalone Book Appointment page discards typed data when patient is new	Medium	Training + V1.1 — train desk on the global modal (which does everything right); fix or retire the standalone page in V1.1.
4. Accepted plan → no "Start Visit" button (tab switch + re-select)	Medium	☐ V1.1 — workaround is 3 clicks; visit gate already covers treatment-type appointments.
5. Visit saved silently; billing prompt discoverable only via tab/badge	Medium	☐ V1.1 (toast/redirect); Week-1 training: desk checks the badge after every chair turnover.
6. No checkout prompt for next appointment / recall	Medium	☐ V1.1 — desk SOP for now; recall engine is the backstop.
7. <code>next_visit_date/type</code> collected, consumed by nothing	Medium	☐ V1.1 — either wire to recall or remove the fields; today it's theatre that erodes trust.
8. Rx button routes to a tab, not the pad (3 screens)	Low	☐ V1.1 .
9. Receipt print not one-click post-payment; no WhatsApp in billing views	Low	☐ V1.1 .
10. No end-of-day close screen (cashbook + income + appointments tour)	Low	☐ V1.1 — cashbook covers the tally; a one-page EOD is the single highest-value V1.1 screen.
11. Huddle CLI writes to a logfile nobody reads	Low	☐ FIX BEFORE SHIP — remove from scheduler or repoint; staff use the web board.
12. Calendar grid doesn't auto-refresh (queue panel only)	Medium	☐ V1.1 — 60-s poll on the grid; meanwhile: two desks share the queue view, not the week view.

SECTION 8 — HUMAN FACTORS

FACTOR	ASSESSMENT
Clicks / screen switching	Money path is tight (modal-based payment, inline plan decisions). Clinical path is the switch-heavy one: 3 screens to consultation, 3 to Rx, 2 tabs plan → visit. A chairside doctor touches ~4-6 screens per patient vs ~2-3 in mature systems.
Learning curve	Low for desk (modals, plain forms, 6 required fields); moderate for doctors (must learn which entry points work — modal vs page, queue URL). The in-app Help Centre is real: 34 KB role-aware content covering 19 screens + workflow guides, with per-route contextual guide component. Use it in Week-1 training.
Fatigue / data-entry burden	Registration burden is minimal (6 fields + skip). Consultation forms are dense but typed variants (Same Issue, Minor Visit, Emergency) exist precisely to shortcut repeat work. The two dead next-visit fields add ~5 wasted seconds per visit — remove or wire (Section 7.7).
Speed	Screens verified eager-loaded and date-bounded; 30-s/60-s polling keeps the desk fresh without websockets. Acceptable on clinic hardware; no heavy JS frameworks.
Chairside usability	Good once inside the patient profile (tabs hold everything); weak getting there from the schedule. The one-glance worksheet with plan items is genuinely chairside-grade.
Reception usability	Strong on booking/registration/payment; the six remembered hand-offs (Section 7) are the fatigue risk — they are exactly the things receptionists forget at 6 PM on a busy day.

SECTION 9 — COMPARISON AGAINST LEADING DENTAL SYSTEMS

Workflow-quality comparison, not feature counting. Benchmarks: Dentrix, Eaglesoft (US legacy desktop); Open Dental (open source); Curve, CareStack, Denticon, Dentally, Sensei Cloud, tab32 (cloud PMS); Cliniko, Practice-Web (lightweight). Assessment from industry knowledge of these products; Dentfluence side is code-verified.

DIMENSION	POSITION	DETAIL
Clinical workflow (plan lifecycle)	AHEAD	The plan=promise / visit=fact / billing-from-visits contract with a single status writer and append-only decision ledger is stricter than any benchmark — Dentrix/Eaglesoft let front desk mutate treatment status freely; Open Dental's plan model is looser. This discipline is Dentfluence's core IP.
Financial workflow (India)	AHEAD	UPI/cash/EMI modes, wallet/advance, GST view, CA export, cashbook — none of the US benchmarks handle Indian clinic accounting; Indian competitors (Clinicia, Bestosys — whose export formats Dentfluence already imports) are far shallower on integrity (no locked numbering, no financial audit trail).
Audit & privacy	AHEAD	Hash-chained audit log with daily tamper verification + field-level PHI encryption exceeds every benchmark listed, including Open Dental.
WhatsApp-first communication	AHEAD (when live)	Consent-gated Cloud API with interactive replies beats the SMS/email stacks of US systems for the Indian market; behind until the dry-run flag flips.
Migration tooling	AHEAD	Named competitor import profiles with annotated column quirks — most PMS vendors make you pay for migration services.
Reception workflow	BEHIND	Dentrix/Curve/Dentally have decades-refined front-desk flow: one-screen check-in → chair → checkout with next-appointment prompt. Dentfluence has the pieces but six manual hand-offs (Section 7).
Appointment book UX	EQUAL-	Drag-drop with conflict guard = parity; no live multi-user grid refresh, no online booking / patient self-scheduling (Dentally, CareStack, Cliniko all have it). Microsite is parked.
Imaging / charting	BEHIND	No X-ray/imaging-device integration and no perio charting — Dentrix, Eaglesoft, Open Dental, CareStack all have both. Tooth chart exists (permanent only; pediatric dentition absent). For an Indian GP practice this is survivable; it is

		the largest clinical feature gap vs benchmarks.
Insurance / claims	N/A-EQUAL	US claims engines don't apply to Tulip's cash/UPI market; no Indian TPA integration either (V2 if corporate patients matter).
Reports	EQUAL-	6-tab reports = solid; misses revenue-per-doctor and production-per-hour that every US PMS leads with.
Mobile	BEHIND	Curve/CareStack/tab32 mobile is mature; Dentfluence's API is competitive but the app is unproven.
AI	EQUAL	Benchmarks ship narrow AI (Overjet-style radiograph reads, CareStack assist). Dentfluence's vision prefill is comparable; Tulip assistant is ahead architecturally, behind on safety — everyone is early here.
Architecture / customization	AHEAD of legacy, EQUAL to cloud	Modern Laravel 13 + API + feature flags beats Dentrux/Eaglesoft desktop tech by a generation; comparable to CareStack/Denticon internally.
Scalability / multi-location	BEHIND	Denticon/CareStack are multi-location natives; Dentfluence's branch scope is deliberately inert and the data plane is single-tenant. Fine for Tulip; the SaaS gap.
Security	AHEAD (post-conditions)	After the 4 P0s close: RBAC + encryption + audit chain + fail-closed webhooks is a stronger posture than the desktop incumbents ever had.

Net position: Dentfluence is already a top-quartile product on *integrity* (money, audit, lifecycle) and India-fit, mid-pack on front-desk polish, and bottom-quartile on imaging and multi-location — exactly the profile of a V1 built by a clinician who cared about the right things first.

SECTION 10 — PRODUCTION RISK MATRIX

LEVEL	RISK	MITIGATION
CRITICAL	No backups (data-loss = practice-loss) · AI RBAC bypass · consultation photos public · failures invisible (no error tracking, mail → log)	The 4 P0s — Days 1-7 of the standing execution plan.
HIGH	Automation silently dead (WhatsApp dry-run + blank creds, mail → log) while staff are told it works · uncommitted Treatment Visit sprint on one disk · cron/queue-worker stop = all 22 jobs stop unalerted · appointments never auto-close → queue statuses wrong by 10 AM	Env flip with staged WhatsApp enable (capped); commit Day 1; uptime ping + Slack channel; today-queue link + Start-Consultation link.
MEDIUM	Communication queue unpaginated over 1,810 backlog · lab/vendor attachments public · missed billing prompts → uncollected revenue · calendar not live for two desks · inventory stockout (no alerts) · VPS outage = total stop (no offline mode) · no revenue-per-doctor for owner	P1 fixes + Week-1 SOPs (badge check per turnover, weekly inventory review, printed morning schedule as downtime fallback).
LOW	next_visit fields unread · Rx/receipt click depth · standalone booking page trap · huddle logfile ritual · payroll is a register only · stale .env.example	V1.1 batch; training notes.
VERY LOW	Race conditions on money/stock (locked) · double-booking (guarded) · CSRF/XSS/SQLi (verified clean) · duplicate patients (guarded) · plan status corruption (single writer)	Already engineered away — the strongest part of the system.

SECTION 11 — V1 FREEZE VALIDATION

MODULE	VERDICT	BASIS
Patients · Consultations · Treatment Plans · Billing/Wallet · Inventory · Prescriptions · Appointments engine · Huddle · Membership · Settings v2 · Marketing/Blog · Reports	FREEZE IMMEDIATELY	Complete, tested, disciplined. Any further development here is risk without return.
Treatment Visits	MINOR FIX ONLY	Commit the sprint, re-run suite, deploy. Nothing else.
WhatsApp	MINOR FIX ONLY	External Meta config + staged enable. Code frozen (CEO already ordered STOP).
PRE / Communication OS	MINOR FIX ONLY	Queue pagination + recall cap; consolidation of the duplicate recall UIs is V1.1.
Lab	MINOR FIX ONLY	Attachments to private disk; one reconciliation smoke test.
HR	MINOR FIX ONLY	Throttle /hr/scan; role_id/role code retirement is V1.1.
Workflow glue (today-queue link, Start-Consultation link, huddle scheduler entry)	MINOR FIX ONLY	Three one-line changes that convert six manual hand-offs into three. Explicitly not redesign.
AI Assistant · Voice · Receipt-scan expansion · ABDM · HQ · Analytics snapshots cutover · Reviews activation · Android write-paths · EOD screen · next-visit wiring	V1.1	Valuable, non-blocking. Sequenced after two clean production weeks.
New consultation types · UI redesigns · marketplace/SaaS features · imaging module · online booking	REJECT (for V1 cycle)	Nothing enters V1. Imaging and online booking are the first V2 conversations, not V1.1.

SECTION 12 — WHAT HAPPENS IF TULIP USES DENTFLUENCE TOMORROW?

HORIZON	PREDICTION (GROUNDED IN THE FINDINGS ABOVE)
Week 1	Registration, booking and payments go smoothly — the desk will like the modals. Pain shows up in status discipline: appointments pile up "in chair" because nothing closes them; 2-3 billing prompts get missed until the badge SOP sticks; a doctor complains it takes three screens to start writing; someone books on the standalone page and loses a half-typed booking; staff phone-WhatsApp because automated reminders are still dry-run. Evening cash tally works first try (cashbook). Expect 5-10 papercut reports — log them, fix only breakages.
Month 1	WhatsApp goes live capped; confirm/cancel replies start updating the queue and reception call-load visibly drops. The recall queue becomes the manager's daily habit — pagination fix proves necessary. First stockout occurs (no inventory alerts) → weekly stock-review SOP is born. Owner asks for revenue-per-doctor (promise it in V1.1). First real restore-drill from backups should happen this month whether needed or not.
Month 3	Data compounds: last_visit_date now accurate for consultation patients, recall targeting improves; visit-only patients gap (P2) becomes measurable → wire it in V1.1. Media storage growth becomes visible on the VPS disk — monitor. Staff fluency makes the six hand-offs invisible; the V1.1 glue batch (EOD screen, start-visit button, toast) lands and Net Promoter jumps more than any feature would.
Month 6	The system is boring — which is the goal. Pressure arrives from success: a second chair/doctor wants per-role logins (BranchScope activation + role_id retirement), corporate patients raise TPA/insurance questions, and Dr. Sumit starts the clinic #2 conversation — which triggers the real V2 gate: clinic_id isolation before any second tenant. DPDP full-lifecycle work (DSAR, breach register) should start now for the May 2027 deadline.

SECTION 13 — FINAL CEO RECOMMENDATION (one page)

Can Dr. Sumit stop development and start implementation? **YES** — after a 7-day hardening pass.
Development stops; implementation starts.

What exactly remains (production blockers only, nothing else):

Days 1-7 — the standing P0/P1 list from audit #2, unchanged: commit the Treatment Visit sprint · backups wired + one restore drill · `ASSISTANT_ENABLED=false` · consultation-photos + lab/vendor attachments to private disk · error visibility (daily logs, Slack alert, real mailer) · production .env hardening incl.

`PRM_WA_APP_SECRET` · verify `patients:encrypt-phi` ran · paginate communication queue · cap recall sends · throttle stray endpoints.

Plus three one-line glue fixes this audit adds (not redesign): link `/appointments/today` in the sidebar · add a "Start Consultation" link on the queue card that passes `appointment_id` (the auto-close observer already exists and is waiting for exactly this) · remove or repoint the logfile huddle scheduler entry.

Plus the config flip that makes the automation real: WhatsApp credentials + staged dry-run-off for appointment reminders only, capped; Brevo SMTP credentials for the briefing emails.

Everything else — EOD screen, start-visit button, Rx shortcut, next-visit wiring, revenue-per-doctor, Android write-paths, Tulip re-enablement, imaging, online booking — is V1.1 or later and must not delay go-live. The system's integrity core (money, audit, lifecycle, privacy) is already better than the incumbents Tulip would be leaving. The remaining weaknesses are training-absorbable friction, not failure modes.

Final readiness estimate: Core Web (Tulip operations): **93%** · Financial engine: **95%** · Automation engine: **82%** (100% coded, held back by config + delivery gaps) · Android companion: **58%** (API 85%, app unverifiable from this repo) · AI assistant: **25%** (disabled for V1 by design) · **Overall Tulip implementation readiness: 90%.** The last 10% is one week of ops work and two weeks of habit — not code.

SECTION 14 — PATH TO "V1 COMPLETE": THE EXECUTION SEQUENCE

Product ladder (locked):

V1 — Fully functional automation unit. No voice, no AI. All workflows and automations live; every role productive; web + Android parity; KPIs complete.

V2 — Voice inputs / transcripts (capture only).

V3 — Processing & suggestions from voice inputs/commands (read-only intelligence).

V4 — AI Secretary (acts on your behalf, confirm-gated writes).

One correction to the V1 definition before the plan — "exact parity" is the wrong target. Screen-for-screen Android parity would take months and deliver zero extra clinic value: nobody bills GST or runs payroll from a phone. The achievable, correct V1 target is **role-based parity**: every role can complete their full daily job on Android — doctor (consult → plan → visit → Rx), assistant, owner (KPIs), manager (huddle, tasks, approvals). Front desk stays desktop-first by design (dense multi-window work). The API already supports this today (~125 write routes, full clinical lifecycle); what remains is client-side validation, not server construction. Accept this definition and V1-complete is ~8 weeks; insist on "exact" and it becomes ~5 months for the same clinical outcome.

Phase A — Week 1 · PRODUCTION HARDENING (go-live gate)

DAY	WORK	EXIT TEST
1	Commit the Treatment Visit sprint (delete stray <code>_treatment-visit-bootstrap.php</code>); run full suite; tag.	Suite green on committed tree
2	Consultation-photos + lab/vendor attachments → private disk via existing secure-media; MIME whitelist on clinical uploads.	Logged-out URL fetch fails
3	<code>ASSISTANT_ENABLED=false</code> ; throttle forgot-PIN/OTP//hr/scan//r/ token; the 3 glue links (sidebar → /appointments/today; "Start Consultation" on queue card passing appointment_id; remove logfile-huddle scheduler entry).	Queue reachable from nav; appointment auto-closes after a linked consultation
4	Backups: nightly <code>backup.sh</code> (DB + storage) in cron, off-server copy, retention 14/60.	Backup exists off-VPS
5	Error visibility: <code>LOG_STACK=daily</code> , Slack channel on critical, real MAIL_MAILER (Brevo creds), uptime ping on /up.	Forced error → alert received; test mail delivered
6	Production .env pass (<code>APP_DEBUG=false</code> , <code>SESSION_SECURE_COOKIE</code> , <code>SESSION_ENCRYPT</code> , <code>PRM_WA_APP_SECRET</code>); run + verify <code>patients:encrypt-phi</code> ; paginate communication queue; recall send cap.	Env checklist signed; queue screen <1s
7	Deploy; re-anchor audit chain; <code>docker compose ps</code> ; <code>dentfluence:smoke</code> ; RBAC walk per role; patient import dry-run with real Clinicia/Bestosys export.	Tulip live on web

Phase B — Week 2 · WORKFLOW GLUE (the six hand-offs → zero)

Ship as one batch of small, independent slices — each is a button/redirect/link on existing plumbing, not redesign. Order: (1) "Start Visit" button on the accepted-plan card (prefill plan_id into the existing openAddForm); (2) visit-save → toast + redirect to the billing prompt; (3) checkout status "Done" → "Book next appointment?" modal reusing the global booking modal; (4) wire `next_visit_date` → creates a recall queue entry (or delete both fields — decide once, this week); (5) consultation Rx button → Rx pad directly; (6) payment success → one-click receipt print + WhatsApp-receipt button reusing the consent-gated link endpoint; (7) fix-or-retire the standalone booking page; (8) 60-second poll on the calendar grid. **Exit test:** a full patient cycle — call → book → arrive → consult → plan → accept → visit → bill → pay → receipt → rebook — with **zero** URL-typing and zero "remember to check" steps.

Phase C — Week 3 · AUTOMATION GOES LIVE

Staged, capped, observable: (1) WhatsApp credentials + Meta template approval → `WHATSAPP_DRY_RUN=false` for **appointment reminders only**, daily cap, watch 3 days; (2) enable confirmations + interactive Confirm/Reschedule/Cancel round-trip; (3) recall sends capped (backlog drains over weeks, not one blast); (4) review requests on; (5) briefing/evening-summary emails confirmed delivered; (6) build the ONE missing automation — a scheduled low-stock alert command reusing `StockStatusService` → AppNotification + huddle (half a day, completes the "automation unit" claim); (7) add a blocked-consent counter to the huddle so staff see who is NOT getting reminders. **Exit test:** every row of the Section 5 table reads Fully/Semi-automatic **and firing**, with delivery visible in-app or in inbox — nothing terminating in a logfile.

Phase D — Weeks 4-6 · ANDROID ROLE-PARITY

The Flutter app leaves "code-complete, untested" only through supervised smoke passes against production API, one role at a time, fixing only breakages: **Week 4 — Doctor:** login/2FA, patient search + profile, consultation create (all types), plan view + present/accept, visit create with worksheet, Rx create + PDF. **Week 5 — Owner + Manager:** dashboard KPIs, reports read, huddle board, task assign/complete, approvals; add the one missing read API this needs (a day-summary finance read endpoint — the only new server code in this phase). **Week 6 — Assistant + payments:** record payment, wallet, inventory lookups, lab status updates; then a full parallel day — one doctor runs Android-only alongside the web desk. **Exit test:** the role-parity checklist (doctor/assistant/owner/manager: every daily task ✓ on Android) is green and each flow was demonstrated on a real device against prod. This lifts Android readiness 58 → ~85.

Phase E — Week 7 · KPI COMPLETION

(1) Revenue-per-doctor in Reports (join payments → invoice items → visits → doctor — the schema already supports it); (2) cut over the analytics snapshot projection (already built, shadow-only — flag on, verify against live numbers for a week); (3) one-page End-of-Day screen: today's collections by mode + pending billing prompts + tomorrow's schedule + cash-in-hand (all four queries already exist in FinanceController/Dashboard — this is composition, not construction); (4) chair-utilization + no-show rate on the owner dashboard. **Exit test:** owner answers "how did we do today, who produced it, what's tomorrow" from two screens (web) and one (Android).

Phase F — Week 8 · V1 FREEZE

Full regression suite + smoke on prod · backup restore drill #2 · Help Centre content updated for the new glue · docs refreshed · tag `v1.0.0` · **declare V1 FROZEN — after this, only defect fixes enter; every idea goes to the V2 ladder.**

The ladder beyond V1 (sequence discipline)

VER	SCOPE	PREREQUISITE THAT MUST COME FROM THE VERSION BEFORE IT
V2	Voice inputs + transcripts — capture only. Local whisper (config already present), voice notes stored on the private disk (PHI note already in config), attached to consultations/visits as documents. No interpretation.	V1's stable clinical records to attach transcripts to; consent framework already live.
V3	Processing & suggestions from voice/commands — read-only intelligence (draft clinical notes from transcript, suggest plan items, flag follow-ups). Human accepts everything.	The per-tool authorization layer — build <code>canAccess()</code> gating into the assistant tool interface FIRST (this is also what un-disables Tulip). V2's transcript corpus to process.
V4	AI Secretary — acts on your behalf: books, reminds, drafts, escalates; confirm-gated writes. The existing Tulip architecture (tools + confirm cards + action log) is already the correct skeleton for this.	V3's proven read-only accuracy + the authorization layer battle-tested + months of V1 workflow data to learn the clinic's patterns from.

Operating rules for the whole sequence (unchanged from your own governance): one slice → stop → report → approval. No phase starts before the previous phase's exit test passes. Nothing skips the ladder — voice before processing, processing before autonomy. And the standing question gates every task: *does this move V1 to production?* If the answer is no, it waits.

PATH TO FINAL V1 — COMPLETE STEP-BY-STEP EXECUTION SEQUENCE

Every step from today's tree to a frozen, fully-operating V1 (web + Android role-parity + all automation live). Steps are the smallest actionable unit — tick the checkbox as each completes (saved in this browser). Rules: work strictly in order inside a phase · a phase's GATE must pass before the next phase starts · Artisan commands are listed for Dr. Sumit to run manually · no step here is a redesign or a new feature beyond the two explicitly marked BUILD items.

PHASE A — Day 1 · Repo hygiene & commit (nothing is safe until it's in git)

✓	#	STEP	WHERE / COMMAND
☐	1	Delete the stray non-blade duplicate file	<code>resources/views/ patients/partials/ _treatment-visit- bootstrap.php</code>
☐	2	Review the 5 modified files (TreatmentVisitController, treatment-visits-tab, patients/show, visits/print, routes/web.php) + 9 untracked (form + 4 partials + TreatmentVisitFormPageTest + 3 report .md)	<code>git status --short</code>
☐	3	Clear compiled views, run the FULL suite, confirm 712 green on this exact tree	<code>php artisan view:clear</code> · <code>php artisan test</code>
☐	4	Commit everything, tag release candidate, push	<code>git add -A && git commit</code> · tag <code>v1.0.0-rc1</code>
☐	5	Check no migration is pending locally vs prod	<code>php artisan migrate:status</code> (local + VPS)
☐	6	Rebuild <code>.env.example</code> to include every required var (WHATSAPP_*, PRM_WA_*, META_*, REVIEWS_*, ANTHROPIC, LOG_SLACK) so no future deploy can miss one	<code>.env.example</code> (currently stock-Laravel only)

GATE A: suite green on committed tree · `git status` clean · tag pushed.

PHASE B — Days 2-3 · Security & privacy fixes (P0-2, P0-3, P1-4, P1-7)

✓	#	STEP	WHERE
☐	7	Repoint <code>consultation-photos</code> disk to the private root (<code>storage/app/private/consultations</code>), remove <code>visibility: public</code>	<code>config/filesystems.php:62-67</code>
☐	8	Serve consultation photos through the existing authenticated route; update any blade referencing <code>/storage/consultations</code>	<code>SecureMediaController · web.php:1171-1174</code>
☐	9	Move existing photo files from public to private path; verify a logged-out URL fetch now fails	one-time move script/command
☐	10	Lab case attachments → private disk + secure serving	<code>LabController.php:555 · LabCaseAttachment.php:51</code>
☐	11	Vendor invoice attachments → private disk	<code>VendorInvoiceService.php:73</code>
☐	12	Add MIME whitelist to clinical upload validation (<code>mimes:jpg,jpeg,png,webp,pdf,mp4,mov,dcm</code> as appropriate)	<code>ClinicalFileController.php:80</code>
☐	13	Disable the AI assistant for V1 + hide its nav/chat entry	<code>ASSISTANT_ENABLED=false · config/assistant.php:19</code>
☐	14	Throttle forgot-PIN + mobile-OTP routes (<code>throttle:5,1</code>)	<code>routes/web.php:45-51</code>
☐	15	Throttle <code>/hr/scan</code> (<code>throttle:30,1</code>) and <code>/r/{token}</code> review pages	<code>web.php:62 · reviews.php:15-16</code>
☐	16	Gate the HR sidebar entry behind <code>canAccess</code> (nav-only leak)	<code>partials/sidebar.blade.php:205</code>

GATE B: logged-out fetch of a consultation photo, lab attachment and vendor invoice all fail · assistant chat gone · 429 on hammered PIN/OTP/scan endpoints.

PHASE C — Days 4-5 · Backups & error visibility (P0-1, P0-4)

✓	#	STEP	WHERE
☐	17	Wire <code>backup.sh</code> (<code>mysqldump</code> + <code>storage/</code>) as a nightly VPS cron (02:30)	<code>repo-root backup.sh</code> → <code>crontab</code>
☐	18	Off-server copy (<code>rclone/scp</code> to second location) + retention 14 daily / 8 weekly	VPS cron
☐	19	Restore drill #1: restore last night's dump to a scratch DB, open a patient, verify encrypted fields decrypt	scratch schema on VPS
☐	20	Switch logging to rotation: <code>LOG_STACK=daily</code>	<code>.env.production</code>
☐	21	Wire the existing slack log channel: <code>LOG_SLACK_WEBHOOK_URL</code> , add slack to the stack for <code>critical</code>	<code>config/logging.php</code>
☐	22	Real mailer: Brevo SMTP credentials (placeholders today), send one test briefing	<code>.env.production:70-78</code>
☐	23	External uptime ping on <code>/up</code> and <code>/api/v1/ping</code> (e.g. UptimeRobot)	free monitor service
☐	24	Cron dead-man switch: append a <code>healthchecks.io</code> ping to the scheduler so a silent cron death alerts you	<code>routes/console.php</code> (one line)
☐	25	Force one error + one critical log → confirm Slack alert arrives	manual test

GATE C: restore drill passed · forced error alerted · test mail delivered · cron dead-man live.

PHASE D — Days 6-7 · Production env, deploy & verification

✓	#	STEP	WHERE / COMMAND
☐	26	<code>APP_ENV=production</code> , <code>APP_DEBUG=false</code>	<code>.env.production</code>
☐	27	<code>SESSION_SECURE_COOKIE=true</code> , <code>SESSION_ENCRYPT=true</code>	<code>.env.production</code>
☐	28	Set <code>PRM_WA_APP_SECRET</code> + <code>PRM_WA_VERIFY_TOKEN</code> (webhook defaults ENABLED — this is its only protection)	<code>config/prm.php:104</code>
☐	29	Fill WhatsApp Cloud creds (<code>PHONE_NUMBER_ID</code> , <code>ACCESS_TOKEN</code> , <code>BUSINESS_ACCOUNT_ID</code>) — keep <code>WHATSAPP_DRY_RUN=true</code>	<code>.env.production:139-143</code>
☐	30	Set <code>REVIEWS_GOOGLE_URL</code> (Google review deep- link)	<code>.env.production</code>
☐	31	Run PHI backfill on prod + spot-check ciphertext in DB	<code>php artisan</code> <code>patients:encrypt-phi</code>
☐	32	Deploy (env changed ⇒ force-recreate)	<code>bash deploy.sh</code> · <code>docker</code> <code>compose up -d --force-</code> <code>recreate</code>
☐	33	Post-deploy ritual: containers up, worker not crash- looping (known stale-timestamp bug)	<code>docker compose ps</code> · <code>php</code> <code>artisan queue:restart</code>
☐	34	Re-anchor the audit hash chain (else <code>audit:verify</code> alarms daily)	per <code>docs/deploy-</code> <code>runbook-2026-07-14.md</code> step 4
☐	35	Money-path smoke on prod (rollback mode)	<code>php artisan</code> <code>dentfluence:smoke</code>
☐	36	RBAC walk: owner / doctor / receptionist logins — no access where none granted; verify appointment- overlap block live on prod	manual, 30 min

GATE D = GO-LIVE READY (web): all P0 closed · smoke green · RBAC verified. Tulip can start supervised daily use after Phase G cutover.

PHASE E — Week 2 · Workflow glue — every hand-off becomes a button

✓	#	STEP	WHERE
☐	37	Sidebar + dashboard link to the live queue <code>/appointments/today</code> (currently orphaned)	<code>sidebar.blade.php</code> · <code>dashboard/index.blade.php</code>
☐	38	Queue card: patient name → patient profile link (cards are dead ends today)	<code>appointments/today.blade.php:134-165</code>
☐	39	Queue card: "Start Consultation" button → <code>patients.consultations.create</code> with <code>?appointment_id=</code>	<code>appointments/today.blade.php</code>
☐	40	Consultation create: pre-select the passed appointment (the observer then auto-closes it + writes <code>last_visit_date</code> — zero new logic)	<code>ConsultationController.php:49-55</code> · <code>consultations/create.blade.php:554</code>
☐	41	Prefill chief complaint from the appointment's complaint field on that same path	same form
☐	42	Accepted-plan card: "Start Visit" button beside "Bill from Plan" → opens visit form with <code>plan_id</code> prefilled (reuse the existing <code>open_visit / openAddForm(prefill)</code> mechanism)	<code>treatment-plan-tab.blade.php:1300</code> · <code>treatment-visit-form-script.blade.php:684-694</code>
☐	43	Visit save: replace silent JSON close with toast "Visit saved — billing prompt created" + inline "Build Invoice now" link to <code>billing.createFromPrompt</code>	<code>TreatmentVisitController::store()</code> · form script
☐	44	Widen the visit gate: fire for consultation-type appointments too, not only <code>type='treatment'</code>	<code>ConsultationController::visitGateData():199</code>
☐	45	Checkout: on status → Done, show "Book next appointment?" prompt that opens the existing global booking modal prefilled with the patient	<code>AppointmentController::updateStatus():307-320</code> + queue JS
☐	46	Decide once: wire <code>next_visit_date/type</code> → creates a <code>communication_queue</code> recall entry on visit save, or delete both fields. (Recommended: wire — it's ~20 lines in <code>TreatmentVisitService</code> .)	<code>TreatmentVisitService.php:72,672</code>
☐			

	47	Consultation "Rx" button → Rx create form directly (not the profile tab)	<code>consultations/show.blade.php:110-112,638</code>
□	48	Payment success: auto-open the new receipt (<code>?auto=1</code> print pattern already exists in Rx) instead of hunting the payment row	<code>billing/show.blade.php:342-379,1170</code>
□	49	WhatsApp button on invoice + receipt views (reuse the consent-gated <code>communication.whatsapp.link</code> endpoint — zero new backend)	<code>resources/views/billing/*</code>
□	50	Standalone booking page: embed the same quick-register partial the global modal uses (controller already accepts first/last/mobile) — or make the page redirect to the modal	<code>appointments/create.blade.php:88,246</code>
□	51	Calendar grid auto-refresh: 60-s poll on the week/day grid (queue panel already polls)	<code>appointments/index.blade.php:2104</code>
□	52	Communication queue: <code>paginate(50)</code> + default status filter + collapse the 4 clone-counts into one grouped query	<code>CommunicationController.php:79-89</code>
□	53	Recall daily send cap (config value, e.g. 40/day) enforced in the send path	<code>RecallEngineService</code> / send command
□	54	Remove (or repoint to mail) the logfile-only <code>tulip:huddle</code> scheduler entry — the web /huddle board is the delivery	<code>routes/console.php:293-296</code>
□	55	Run full suite again; add small feature tests for steps 39-46; deploy	<code>php artisan test</code> · <code>bash deploy.sh</code>

GATE E: one full patient cycle (call → book → arrive → consult → plan → accept → visit → bill → pay → receipt → rebook) completed by a receptionist + doctor with ZERO url-typing, zero tab-hunting, zero "remember to" steps.

PHASE F — Week 3 · Automation goes live (staged, capped, observable)

✓	#	STEP	WHERE
☐	56	Submit <code>appointment_reminder</code> template (with Confirm/Reschedule/Cancel buttons) for Meta approval; verify all 7 templates approved	Meta Business Manager · <code>config/whatsapp.php</code>
☐	57	Webhook GET handshake on prod (verify token) + bad-HMAC POST rejected with 401	<code>/api/v1/webhooks/prm/whatsapp</code>
☐	58	Dry-run E2E: run <code>whatsapp:send-reminders</code> , inspect payloads + consent-blocked count in the log	<code>storage/logs/whatsapp-reminders.log</code>
☐	59	Flip <code>WHATSAPP_DRY_RUN=false</code> — reminders only , daily cap, staff phones as first recipients; watch 3 days	<code>.env.production</code> + <code>--force-recreate</code>
☐	60	Live interactive round-trip: patient taps Confirm/Cancel → queue status updates / cancel executes	<code>InboundMessageService:122-165</code>
☐	61	Enable appointment confirmations on booking	template <code>appointment_confirmation</code>
☐	62	Enable recall sends under the Phase-E cap — the 1,810 backlog drains over ~6 weeks, never as a blast	recall pipeline
☐	63	Enable review requests (<code>reviews:request 11:00</code> ; one-per-appointment guard already exists)	<code>REVIEWS_ENABLED</code> + google URL
☐	64	Confirm morning briefing (07:05), SLA alert (14:00), evening summary (18:00) arrive in real inboxes	Brevo delivery logs
☐	65	BUILD (half-day): <code>inventory:scan-low-stock</code> daily 07:25 — reuse <code>StockStatusService</code> , AppNotification to manager + huddle line. The one missing automation.	new command + <code>routes/console.php</code>
☐	66	BUILD (small): blocked-consent counter on the huddle ("N patients not receiving reminders — no WhatsApp consent") so silence is visible	<code>HuddleService</code>
☐	67	Extend <code>last_visit_date</code> writer to treatment visits (observer or service hook, monotonic like the consultation	<code>ConsultationClinicalWiringObserver</code> pattern → <code>TreatmentVisitService</code>

		one) — recall accuracy for visit-only patients	
□	68	Clean orphan relationship rows before recall volume rises	<code>php artisan relationship:backfill --apply</code>

GATE F: every automation row in Section 5 firing AND delivering somewhere a human looks (in-app, inbox, or WhatsApp) — nothing terminating in a logfile.

PHASE G — Cutover · Data migration & go-live at Tulip (overlaps Phase E week)

✓	#	STEP	WHERE
☐	69	Verify Settings → Data tab actually renders the import form (flagged unverified after an 08-03 cleanup)	<code>settings.index?tab=data</code>
☐	70	Export full patient base from the old software (Clinicia/Bestosys export)	old system
☐	71	Import preview → check the first-10 mapping + duplicate count → import (chunked 500/transaction; legacy UHID preserved as patient_id)	<code>PatientImportExportController</code>
☐	72	Spot-check 20 imported patients incl. phone-duplicate families	manual
☐	73	Masters setup: branches, doctors, operatories, treatment categories + pricing, consent purposes (seeder), Rx drug master, letterheads/print templates	Clinic Administration
☐	74	Staff accounts: one per human (no shared logins), owner-configured roles + module permissions matrix, 2FA on admin/owner accounts	Roles & Permissions
☐	75	Opening balances: outstanding patient dues entered as opening invoices; advances as wallet deposits (SOP: one line item "Opening balance", dated cutover day)	Billing / Wallet
☐	76	Re-enter ACTIVE treatment plans only (patients mid-treatment) + next 2 weeks of appointments	manual, front desk + doctor
☐	77	Inventory opening stock: product bulk import + opening stock-count session	<code>InventoryProductImportController</code> · Stock Count
☐	78	Training half-day 1 (desk): global booking modal ONLY, today-queue screen, billing-badge SOP, payment modal, cashbook EOD, wa.me sends	Help Centre as courseware
☐	79	Training half-day 2 (doctors): queue → Start Consultation, typed consultation variants, plan present/accept chairside, visit worksheet, Rx	Help Centre
☐	80	Downtime SOP printed & posted: morning schedule print, paper fallback forms, who to call (no offline mode exists)	one A4 sheet
☐			

	81	Parallel day: old system READ-ONLY, Dentfluence primary; evening compare collections both systems	1 clinic day
□	82	GO-LIVE DAY (supervised): old software off; papercut log open; evening review; backup confirmed that night	—
□	83	Week-1 rhythm: daily 15-min papercut review; fix ONLY breakages; everything else → V1.1 register	—

GATE G: Tulip runs a full week on Dentfluence alone · evening cash tally matches physically · zero data-loss incidents.

PHASE H — Weeks 4-6 · Android role-parity (module by module, fix-only)

✓	#	STEP	SCOPE
☐	84	Build + distribute current APK; device matrix (doctor's phone + one spare); point at prod API	Flutter repo · flutter pub get first (known gotcha)
☐	85	Smoke: login, 2FA, token refresh, logout-all	/api/v1/auth/*
☐	86	Doctor read paths: patient search, profile, timeline, clinical files view	Patient/PatientProfile controllers
☐	87	Consultation create — all 4 typed variants + COHA — against prod	POST /patients/{p}/consultations*
☐	88	Treatment plan: view, mark-presented, accept/defer/reject, partial-accept	/treatment-plans/{plan}/*
☐	89	Visit create with worksheet from plan items; edit; verify billing prompt appears on web after mobile visit	POST /patients/{p}/visits
☐	90	Prescription: create, finalize, repeat, PDF; drug alerts endpoints	/prescriptions* + /rx/*
☐	91	Clinical photo upload from phone camera → lands on PRIVATE disk → visible on web	POST /patients/{p}/clinical-files
☐	92	Appointments: book, walk-in, reschedule, status updates from phone	/appointments*
☐	93	Payments: record payment (all modes), wallet credit, EMI installment; verify receipt on web	POST /invoices/{i}/payments
☐	94	Owner: dashboard KPIs + 6 report tabs read on mobile	Dashboard/Report controllers (read-only)
☐	95	BUILD (1 day, only new server code in this phase): finance day-summary read endpoint (collections by mode + cash-in-hand) for the owner app view	Api/V1 — new read route
☐	96	Manager: huddle board, task assign/complete, notifications on mobile	Huddle API
☐	97	Assistant: inventory lookups, stock status, lab case status transitions	Inventory/Lab API
☐	98	Membership regression (the one already-tested module)	Membership API
☐	99	Full parallel day: one doctor works Android-ONLY beside the web desk; log every stumble; fix only breakages	1 clinic day
☐	100	Sign the role-parity checklist (doctor / assistant / owner / manager: every daily task ✓ on Android)	this document

GATE H: parity checklist signed on real devices against prod. Android 58 → ~85.

PHASE I — Week 7 · KPI completion + Week 8 · FREEZE

✓	#	STEP	WHERE
☐	101	BUILD: revenue-per-doctor in Reports (payments → invoice items → visits → doctor join; schema supports it today)	<code>ReportsController::buildRevenueData(): 417</code>
☐	102	BUILD: one-page End-of-Day screen — today's collections by mode, pending billing prompts, cash-in-hand, tomorrow's schedule (all 4 queries already exist; composition only)	new route + view over <code>FinanceController / DashboardController</code>
☐	103	Analytics snapshot cutover: flag ON, verify against live numbers for one week, then retire the shadow note	<code>RebuildAnalyticsSnapshots</code> + feature flag
☐	104	Chair-utilization % + no-show rate cards on owner dashboard (queue page already computes utilization — reuse)	<code>appointments/today.blade.php</code> logic → dashboard
☐	105	Full regression suite + prod smoke + restore drill #2	<code>php artisan test · dentfluence:smoke</code>
☐	106	Update Help Centre content for the new glue (queue buttons, EOD screen); refresh deploy runbook	<code>resources/help/content.php</code> · <code>docs/</code>
☐	107	Tag <code>v1.0.0</code> · DECLARE V1 FROZEN · open the V1.1/V2 register — after this, only defect fixes enter V1	<code>git tag v1.0.0</code>

GATE I = V1 COMPLETE: all workflows buttoned, all automations live and observed, role-parity on Android signed, owner KPIs answerable from two screens, system frozen. Next conversation is V2 (voice capture).

Explicitly NOT in this path (so nobody sneaks them in): AI assistant re-enable (V1.1 — needs per-tool canAccess first) · voice/transcripts (V2) · imaging/perio charting · online booking/microsite · clinic_id multi-tenant · GBP/Meta integrations · UI redesigns · payroll engine · SMS channel · Reception-parity Android. Every one of these delays V1 and none of them is needed for Tulip to operate.

Prepared 2026-08-07 (third independent audit) · Basis: two full workflow-tracing passes across blade views, controllers, services, observers, scheduler, routes and API, spot-checked; prior audits 2026-08-03 and 2026-08-07 AM; governance record 2026-05 → 2026-08. Flutter app source is not present in this repository — Android conclusions are API-contract-based and marked accordingly. Verdict: **YES WITH CONDITIONS — stop development, start implementation after the 7-day hardening pass. Overall CEO Score: 84/100. Implementation readiness: 90%.**